PassFrame tuition

Coordinates & Transformations

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: A point is at (-4, 3). It is reflected in the y-axis, and the result is then translated 2 down. What are the final coordinates?

Method & answer: Reflect in y-axis: (4, 3). Translate 2 down: (4, 1).

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point A is at (-6, 2). What is its x-coordinate?

2. A point A is at (-6, 2). What is its y-coordinate?

3. A point starts at (-3, 5) and is translated 6 right. What are its new coordinates?

4. A point starts at (2, -7) and is translated 4 up. What are its new coordinates?

5. A point is at (-2, -4). It is translated 5 right. What are its new coordinates? A) (3, -4) B) (-7, -4) C) (-2, 1) D) (-2, -9)

6. A point is at (-1, 5). Which quadrant of the grid is it in?

7. A point is at (4, -6). Which quadrant of the grid is it in?

8. A point starts at (-8, 1) and is translated 3 right and 4 down. What are its new coordinates?

9. A point starts at (5, 6) and is translated 9 left and 3 down. What are its new coordinates?

10. Select ALL that apply. Look at the points P(-3, 5), Q(3, -5), R(-3, -5) and S(3, 5). Which do NOT lie in the second quadrant? A) P(-3, 5) B) Q(3, -5) C) R(-3, -5) D) S(3, 5)

11. A point starts at (0, -8) and is translated 4 left and 4 up. What are its new coordinates?

12. A point starts at (-7, -3) and is translated 9 right and 8 up. What are its new coordinates? A) (2, 5) B) (-16, -11) C) (2, -11) D) (-16, 5)

13. A point is at (-8, 5). It is reflected in the x-axis. What are its new coordinates?

14. A point starts at (6, -2) and is translated 10 left and 6 up. What are its new coordinates?

15. Which of these points lies exactly on the x-axis? A) (-8, 0) B) (0, -8) C) (8, 8) D) (-8, -8)

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point is at (-5, 6). It is reflected in the x-axis. What are its new coordinates?

2. A point is at (7, -4). It is reflected in the y-axis. What are its new coordinates?

6. A point is at (-9, -5). It is reflected in the y-axis. What are its new coordinates?

7. A line segment joins A(-5, 2) and B(5, 2). What is the length of AB?

11. Select ALL that apply. A rectangle has vertices at (-4, -3), (6, -3), (6, 5) and (-4, 5). Which statements are true? A) Its width is 10 units B) Its height is 8 units C) The vertex (6, 5) is diagonally opposite (-4, -3) D) All four vertices have a negative x-coordinate

12. A point starts at (-2, -2) and is translated 11 right and 11 up. What are its new coordinates?

3. A point starts at (-6, -6) and is translated 10 right and 10 up. What are its new coordinates?

4. A point at (-8, -3) is reflected in the x-axis. What are its new coordinates? A) (-8, 3) B) (8, -3) C) (8, 3) D) (-8, -3)

5. A point is at (6, 5). It is reflected in the x-axis, then the result is reflected in the y-axis. What are the final coordinates?

8. A point starts at (10, -8) and is translated 13 left and 6 up. What are its new coordinates?

9. A point at (8, -6) is reflected in the y-axis. What are its new coordinates? A) (-8, -6) B) (8, 6) C) (-8, 6) D) (6, -8)

10. A point is at (-10, -7). It is reflected in the x-axis. What are its new coordinates?

13. A point at (-3, 7) is reflected in the x-axis to give P, then P is translated 4 down. What are the final coordinates?

14. A point starts at (-6, 7) and is translated 9 right and 10 down. What are its new coordinates? A) (3, -3) B) (-15, -3) C) (3, 17) D) (-15, 17)

15. A point starts at (0, 0) and is translated 8 left and 8 down. What are its new coordinates?

Standard / Advanced — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. A point at (3, -6) is rotated 90° clockwise about the origin, then the result is translated 2 left. What are the final coordinates?

2. A point starts at (-5, -4). What is its distance from the origin along the x-axis alone (i.e. how far would it need to travel right to reach x = 0)?

3. A rectangle has vertices at (-4, -2), (6, -2), (6, 3) and (-4, 3). What is its perimeter?

4. A point is at (-2, -2). It is reflected in the x-axis to give point P. What is the sum of the x-coordinate and y-coordinate of P?

5. A point at (8, -3) is translated 8 left, landing exactly on which axis?

6. A point is at (-4, 7). It is rotated 90° clockwise about the origin. What are its new coordinates?

7. A square has vertices at (-3, -3), (3, -3), (3, 3) and (-3, 3). What is its side length, and what are the coordinates of its centre?

8. A point is at (-6, -2). It is rotated 90° anticlockwise about the origin. What are its new coordinates?

9. A point is at (-9, 8). It is reflected in the y-axis. What are its new coordinates?

10. Select ALL that apply. A shape has vertices A(-2, 0), B(-2, 5), C(4, 5) and D(4, 0). It is rotated 180° about the origin. Which of these are correct new vertices? A) A becomes (0, -2) B) B becomes (2, -5) C) C becomes (-4, -5) D) D becomes (-4, 0)

11. A point starts at (7, -10) and is translated 15 left and 16 up. What are its new coordinates?

12. A point at (5, -8) is reflected in the y-axis, and the result is then rotated 90° clockwise about the origin using (x, y) → (y, -x). What are the final coordinates?

13. A rectangle has vertices at (-6, -4), (2, -4), (2, 3) and (-6, 3). What is its perimeter? A) 30 units B) 15 units C) 24 units D) 38 units

14. A point is at (-2, -11). It is reflected in the x-axis. What are its new coordinates?

15. A point starts at (-8, -8) and is translated so that it ends up at (0, 0). What was the translation?